

ABU BAKAR SHABBIR

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SUMMARY

Innovative Software Engineer with strong expertise in Machine Learning, Deep Learning, and NLP. Proficient in Python, C++, Django, and SQL with experience in developing data-driven and research-oriented solutions. Skilled in applying AI and machine learning techniques to solve real-world problems. Passionate about building intelligent, reliable, and high-performance software systems.

WORK EXPERIENCE

Research Assistant – PhD Coursework

2025 – Present

University of Management & Technology

Supported PhD-level research activities through comprehensive literature reviews, critical analysis of academic studies, and contributions to research design and methodology development. Collaborated closely with faculty on research planning, synthesis of findings, and preparation of academic materials and documentation.

Teaching Assistant (OOP)

Oct 2024 – Mar 2025

University of Management & Technology

Mentored and assisted in grading 200+ students in Object-Oriented Programming; strengthened conceptual understanding through targeted guidance and co-developed tutorials and coding exercises to enhance course delivery.

Peer Tutor (Programming, OOP & Data Structures)

Jul 2024 – Aug 2025

University of Management & Technology

Conducted 50+ tutoring sessions in Programming Fundamentals, OOP, and Data Structures (C++), improving student performance through personalized, hands-on instruction.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, SQL, JavaScript, HTML/CSS

Machine Learning & AI: Machine Learning, Deep Learning, NLP (TensorFlow, PyTorch, Scikit-learn)

Data Science & Research: Data Analysis, Data Visualization, Research Methodology, Academic Research

Frameworks/Tools: Django, Bootstrap, Git, GitHub, Qt Designer (Desktop Application Development)

Core: Data Structures & Algorithms, Object-Oriented Programming (OOP), Design Patterns, Model Development & Deployment, Software Development

Databases: MySQL, SQLite

EDUCATION

2023 – Present	BS in Software Engineering , University of Management & Technology, Lahore, Pakistan	CGPA: 3.66/4.0
2021 – 2023	FSc (Pre-Engineering) , Board of Intermediate and Secondary Education, Sahiwal, Pakistan	
2019 – 2021	Matriculation (Science) , Board of Intermediate and Secondary Education, Sahiwal, Pakistan	

CERTIFICATIONS & ACHIEVEMENTS

- **IBM Data Science Professional Certificate** – Completed comprehensive data science training including Python, SQL, Machine Learning, and Data Visualization.
- **Machine Learning Specialization-Stanford University** – Covered supervised/unsupervised learning, neural networks, and deployment pipelines.
- **Understanding Research Methods-University of London** – Focused on research design, methodology, and academic writing.
- **Microsoft Excel Certification** – Demonstrated skills in data analysis, formulas, and visualization.
- **Peer Tutor Certification (Fall & Spring 2024)-UMT** – Recognized for mentoring 50+ students in programming and data structures.
- **Teaching Assistant – Object-Oriented Programming-UMT** – Assisted in grading and mentoring 200+ students, enhancing learning outcomes.
- **Letter of Recommendation – Programming-UMT** – Endorsed by faculty for excellence in tutoring and teaching support.

PROJECTS

Bank Application	Django banking app for accounts, transactions, loans, and mobile recharge. Implements Strategy, Facade, and Factory patterns for scalable and secure operations. [GitHub]
Election Management System	Secure web-based voting system with OTP auth, role-based access, and real-time result visualization. [GitHub]
Skin Disease Classification (CNN)	CNN-based skin disease detection with high reliability for scalable dermatology diagnostics. [GitHub]
Apple Plant Disease Detection	CNN model detecting apple leaf diseases for precision agriculture and crop health monitoring. [GitHub]
Gender Classification (Deep Learning)	CNN-based gender classification from facial images; demonstrates robust computer vision applications. [GitHub]
CKD Detection Using ML	Early detection of Chronic Kidney Disease using ML classifiers; supports preventive healthcare. [GitHub]
Cardiovascular Risk Prediction	Predicts heart disease risk using clinical data; compares multiple classifiers for personalized care. [GitHub]
Medicine Price Prediction	Predicts medicine prices using multiple regression models; supports data-driven pricing and cost management. [GitHub]
Length of Stay Prediction	Forecasts hospital stay duration for resource planning; compares raw vs. preprocessed datasets for improved prediction. [GitHub]
Hospital Stay Prediction	Predicts patient stay using hospital infrastructure data; aids admins in planning and operational decisions. [GitHub]
Liver Disease Risk Prediction	Early detection of liver disease using clinical and biochemical features; improves predictive accuracy. [GitHub]
Diabetes Outcome Prediction	Predicts diabetes risk using clinical features; highlights improvements with data preprocessing. [GitHub]
Brain Tumor Classification	MRI-based brain tumor classification for early diagnosis and treatment planning. [GitHub]

PUBLICATIONS

- SHABBIR, A. B., Husnain, U. (2026). *Towards Robust and Explainable Early Rumor Detection: A Noise-Aware Graph Learning Framework* UMT Artificial Intelligence Review, 5(2), 1-22. <https://doi.org/10.32350/umt-air.52.01>.
- Shabbir, A. B. *SHAP-Weighted Hybrid Ensemble for Interpretable Stroke Risk Prediction Using Tabular Clinical Data*. Robotics and Artificial Intelligence Review, 2025.
- Shabbir, A. B. *Towards a Software Engineering Framework for Health-Aware AI in Women's Digital Work Platforms*. Foundation University Journal of Engineering and Applied Sciences (FUJEAS), 2026, [Under Review](#).
- Shabbir, A. B. *Multi-Class Depression Detection from Social Media Integrating Machine Learning, LSTM, and BERT Approaches for Mental Health Analysis*. Innovative Computing Review, 2026, [Under Review](#).
- Shabbir, A. B. *A Review of Federated Learning in Medical Imaging: Techniques, Challenges, and Future Directions*. 2026, [Under Review](#).
- Shabbir, A. B. *Self-Supervised Learning and Foundation Models in Medical Imaging: A Comparative Review of Methods and Applications*. Journal of Computer & Intelligent System, 2026, [Under Review](#).
- Shabbir, A. B. *Explainable Hybrid CNN-Attention Network for Severity-Based Dysgraphia Detection from Handwritten Samples*. 2026, [Under Review](#).
- Shabbir, A. B. *Explainable and Fairness-Aware Machine Learning for Early Autism Risk Screening Using Behavioral and Demographic Data*. 2026, [Under Review](#).
- Shabbir, A. B. *Explainable Multi-Task Vision Transformer for Automated Autism Spectrum Disorder Detection from Facial Images*. 2026, [Under Review](#).